THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A real image can be formed on a: 1

(a) wall of glass only **(b)** screen
(c) air **(d)** mirror only

2. The mirror that diverges light is: 1

(a) concave **(b)** convex
(c) plane **(d)** cylindrical glass

3. A periscope has: 1

(a) one convex mirror **(b)** a prism only
(c) two lenses only **(d)** two plane mirrors

4. Lateral inversion means: 1

(a) left-right reversal **(b)** shrinking
(c) up-down reversal **(d)** enlargement

5. A rear-view mirror is: 1

(a) convex **(b)** cylindrical **(c)** plane **(d)** concave

6. The number of images by two mirrors at 90 degrees is: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 1

7. Light travels in: 1

(a) zigzag **(b)** spirals **(c)** circles **(d)** straight lines

8. A shaving mirror is: 1

(a) concave **(b)** convex
(c) plane **(d)** cylindrical glass

9. A kaleidoscope forms patterns by: 1

(a) refraction **(b)** diffraction
(c) dispersion **(d)** multiple reflections

10. Concave mirror is used by: 1

(a) dentist **(b)** photographer only
(c) in periscope **(d)** driver as rear-view

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is lateral inversion? 2

12. Name two types of reflection. 2

13. What is the focal length? 2

14. What is a kaleidoscope? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If i = 30 degrees, find r and the angle between the incident and reflected rays. 3

16. Name three optical devices that use lenses. 3

17. Why does the vehicle's number appear correct in a mirror when written 'inverted' on the front? 3

OR

If a ray falls at 0 degrees on a mirror, find i and r. 3

18. Write three differences between convex and concave lenses. 3

OR

Name the parts of a concave mirror. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how a periscope works. 5

OR

Describe an activity to find the focal length using the Sun. 5

20. Describe convex and concave lenses with uses. 5

OR

Explain how spectacles correct short and long sight. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND